

Dynamic Guideline: WHO-Based Measles - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Scope note and WHO guideline set synthesized

This report synthesizes **exactly 5 WHO measles-related guidance documents** relevant to diagnosis, treatment/clinical management, outbreak response, vaccination strategy, surveillance, and elimination verification. The most recent formal WHO measles-related publication identified in the source set is the **14 January 2025 interim guidance on targeted and selective strategies in measles and rubella vaccination campaigns** listed on WHO's measles topic page [WHO measles topic page](#). The most recent WHO publication matching the requested title pattern, however, is **not measles-related: "Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)"**, published **11 December 2025**; it is a case study of the WHO abortion care guideline, not a measles guideline [WHO Western Pacific publication page](#).

No.	WHO guidance document synthesized	Date / update identified	Population / setting	Scope and objectives
1	Targeted and selective strategies in measles and rubella vaccination campaigns: Interim guidance	14 January 2025	Countries planning measles/rubella vaccination campaigns	Supports targeted/selective vaccination strategies and operational planning toward measles/rubella objectives WHO measles topic page .
2	Measles Outbreak Guide	2022	All settings striving for or approaching measles elimination	Updates protocols for measles surveillance, outbreak investigation, response, and recovery; developed by WHO and partners WHO Measles Outbreak Guide .
3	WHO Guide for clinical case management and infection prevention and control during a measles outbreak	2020	Health facilities and outbreak-response teams	WHO outbreak toolbox lists it as a key reference for clinical case management and infection prevention and control during measles outbreaks WHO Measles Outbreak Toolbox .
4	Guidelines on Verification of Measles and Rubella	2019	National verification committees, ministries of health, WHO	Provides verification criteria, lines of evidence, surveillance and vaccination-coverage

No.	WHO guidance document synthesized	Date / update identified	Population / setting	Scope and objectives
	Elimination in the Western Pacific Region – Second Edition		Western Pacific countries/areas	documentation, and case/source classification for elimination verification WHO IRIS , WHO IRIS PDF .
5	Measles — WHO Vaccine-Preventable Diseases Surveillance Standards	Last updated 15 October 2018	Surveillance programmes and public-health authorities	Defines clinical and surveillance features, incubation, rash timing, infectious period, and surveillance standards for measles WHO Surveillance Standards PDF .

Key related WHO supporting documents not counted among the required 5-guideline set:

- **Measles vaccines: WHO position paper – April 2017**, which replaced the 2009 position paper and includes WHO/SAGE vaccine recommendations and GRADE evidence materials [WHO position paper page](#), [WHO IVB measles page](#).
- WHO laboratory-surveillance manual for measles/rubella/CRS, which provides protocols for diagnosis and molecular characterization [WHO manual page](#).

1.2 Key Recommendations

GRADE interpretation used in this report

GRADE domain	Interpretation
High	Further research is unlikely to change confidence in the effect estimate or operational recommendation.
Moderate	Further research may affect confidence and could change the estimate or operational details.
Low	Further research is likely to affect confidence and may change the estimate.
Very Low	Evidence is uncertain; recommendation is mainly based on indirect evidence, expert consensus, or operational necessity.
Strong recommendation	Benefits clearly outweigh harms/burdens, or the recommendation is essential for public-health control.
Conditional recommendation	Applies to selected populations, contexts, risk groups, or resource settings.

Recommendation	Supporting evidence / rationale	GRADE Evidence Quality	Strength
Maintain very high two-dose measles-containing vaccine coverage, operationally targeting ≥95% coverage with two timely doses.	WHO IA2030 states that because measles is highly contagious, 95% coverage with two timely doses is required to prevent spread WHO IA2030 PDF . WHO 2017 vaccine evidence reports median effectiveness of 94.1% for two doses versus no vaccination WHO references PDF .	High	Strong
Use targeted or selective measles/rubella vaccination campaign strategies where appropriate to close immunity gaps.	WHO lists 2025 interim guidance on targeted/selective measles-rubella vaccination campaigns to support countries working toward measles objectives WHO measles topic page .	Moderate	Conditional
Rapidly identify suspected measles using fever plus generalized maculopapular non-vesicular rash, or clinician suspicion; clinical measles includes rash plus cough, coryza, or conjunctivitis.	WHO outbreak toolbox provides suspected and clinical case definitions WHO Measles Outbreak Toolbox .	Moderate	Strong
Confirm suspected measles with laboratory testing and molecular surveillance where feasible.	WHO laboratory manual provides tools and protocols for diagnosis and molecular characterization WHO manual page . CDC recommends measles-specific IgM and measles RNA by RT-PCR for confirmation CDC infection control: measles .	Moderate	Strong
Apply airborne precautions and isolate measles cases through 4 days after rash onset.	WHO surveillance standards and CDC guidance state infectiousness usually extends from about 4 days before to 4 days after rash onset WHO Surveillance Standards PDF , CDC clinical overview .	Moderate	Strong
Manage uncomplicated measles supportively; monitor and treat complications.	Recent reviews confirm no established antiviral therapy is approved for routine measles	Moderate	Strong

Recommendation	Supporting evidence / rationale	GRADE Evidence Quality	Strength
	treatment; management remains supportive PMC updated literature review .		
Administer vitamin A to children with measles, especially severe disease, hospitalization, malnutrition, or vitamin A deficiency risk; use age-based dosing under clinician supervision.	WHO-referenced dosing: 50,000 IU for <6 months, 100,000 IU for 6–11 months, 200,000 IU for >1 year, given for 2 doses; third dose if deficiency signs PMC review , Medscape . CDC emphasizes vitamin A does not prevent measles and is not a vaccine substitute CDC clinical overview .	Moderate	Strong for severe/high-risk children; Conditional for universal use in high-income settings
Offer post-exposure prophylaxis to susceptible exposed persons: MMR within 72 hours if eligible; immune globulin within 6 days for selected high-risk groups or when vaccine is contraindicated.	CDC, ACOG, NACI, and systematic review data support MMR within 72 hours and immune globulin within 6 days; MMR and immune globulin should not be given simultaneously CDC vaccine recommendations , ACOG, NACI recommendations , PubMed PEP systematic review .	Low to Moderate	Strong
During outbreaks, prioritize rapid case investigation, contact tracing, vaccination, PEP, isolation/quarantine, surveillance strengthening, and stock management for vaccines and supplies.	WHO Outbreak Guide addresses surveillance, investigation, response, and recovery WHO Measles Outbreak Guide . WHO 2025 Disease Outbreak News recommends rapid response teams, strengthened border-area surveillance, preparedness, and vaccine/supply stocks WHO Disease Outbreak News .	Moderate	Strong
Verify elimination only after at least 36 months of interrupted endemic transmission under high-quality surveillance.	WHO Western Pacific and Eastern Mediterranean documents use the 36-month interruption standard for verification WHO IRIS PDF , WHO EMRO PDF .	Low for direct trial evidence; High for global operational standard	Strong
Classify cases by final classification and source of	WHO WPR guidance specifies final classifications—laboratory confirmed, epidemiologically	Moderate	Strong

Recommendation	Supporting evidence / rationale	GRADE Evidence Quality	Strength
infection for elimination documentation.	linked confirmed, clinically compatible, discarded—and source categories—imported, import-related, endemic, unknown WHO IRIS PDF .		

1.3 Guideline Development Methodology

Evidence Review Process

- **WHO vaccine policy:** The 2017 WHO measles vaccine position paper was informed by SAGE discussions in **November 2013, October 2015, and October 2016** and is accompanied by WHO GRADE evidence documents on effectiveness, duration of protection, HIV, safety, and revaccination [WHO IVB measles page](#), [NITAG Resource Center](#).
- **Outbreak guidance:** The 2022 WHO Measles Outbreak Guide was developed by WHO and partners and draws on existing WHO documents on outbreak response, case management, infection prevention, surveillance standards, laboratory surveillance, vaccine policy, and outbreak recovery [WHO Measles Outbreak Guide](#).
- **Surveillance and verification:** WHO verification guidance uses standardized criteria and lines of evidence, including surveillance quality, epidemiology, molecular data, vaccination coverage, and population immunity documentation [WHO IRIS PDF](#), [WHO EMRO PDF](#).

Consensus Method

- **SAGE-based vaccine recommendations:** WHO measles vaccine recommendations were discussed through SAGE, WHO's principal advisory group for immunization policy [NITAG Resource Center](#).
- **Verification governance:** Countries are expected to establish **National Verification Committees**, while WHO regions use **Regional Verification Commissions** to review elimination evidence [WHO EMRO PDF](#), [WHO South-East Asia Regional Verification Commission](#).
- **Western Pacific verification process:** The WPR second edition requires documentation against verification criteria and lines of evidence, with subnational analysis of epidemiology, surveillance performance, and vaccination coverage [WHO IRIS PDF](#).

Conflict of Interest Management

- For the measles-specific sources available here, explicit conflict-of-interest details were not visible in the retrieved summaries for each document. The WHO vaccine position paper process is linked to SAGE and GRADE evidence materials, which normally include structured evidence review and recommendation processes [WHO IVB measles page](#).

II. Evidence Summary After WHO Latest Guideline Release (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

Evidence was synthesized from WHO technical documents, WHO outbreak updates, CDC clinical and surveillance guidance, CDC MMWR, PubMed-indexed systematic reviews and reviews, national advisory guidance, and selected authoritative clinical guidance published or updated through **29 April 2026**.

Databases and Sources Searched

- WHO official publications, WHO IRIS, WHO outbreak toolbox, WHO Disease Outbreak News
- CDC measles clinical overview, vaccine recommendations, infection-control guidance, MMWR, case surveillance
- PubMed-indexed systematic reviews and clinical reviews
- ACOG, NACI, AAD, Emergency Care BC, Medscape clinical references
- Measles & Rubella Partnership publication lists

Inclusion Criteria

- Measles diagnosis, clinical management, infection prevention, vaccination, post-exposure prophylaxis, outbreak control, surveillance, or elimination verification
- WHO guidance and authoritative public-health guidance
- Systematic reviews, reviews, surveillance updates, and outbreak reports from 2024–2026 where available
- Evidence available up to **29 April 2026**

Exclusion Criteria

- Non-measles guidance, except where needed to clarify that the requested youth-engagement WHO case study is not measles-related
- Unverifiable claims or sources without accessible title/date information
- Animal-only or preclinical-only evidence unless directly relevant to emerging antiviral research

2.2 Key New Studies

Study / source	Design	Key findings	GRADE Quality
Acute management of measles: A systematic review of therapeutic strategies , <i>Antiviral Research</i> , 2026	Systematic review	Vitamin A remains the only widely recommended acute measles therapy; ribavirin and interferon-α have been used off-label with evidence limited to small trials and case reports PubMed .	Moderate for vitamin A; Low for antivirals
Post-exposure prophylaxis for the prevention of measles , 2025	Systematic review	Included 9 studies and 660 PEP recipients; Ig PEP effectiveness ranged 76%–100% , MCV/MMR PEP 83.4%–100% , but studies had risk-	Low to Moderate

Study / source	Design	Key findings	GRADE Quality
		of-bias and reporting limitations PubMed .	
Updated measles literature review , 2026	Narrative/review article	No antiviral therapy is established for measles; prevention remains the mainstay. More trials are needed for rapid antiviral therapy in outbreak settings PMC updated literature review .	Low to Moderate
CDC Measles Update — United States, Jan. 1–Apr. 17, 2025 , MMWR	Surveillance/outbreak report	Measles control depends on vaccination, isolation, quarantine, and PEP; increasing national and local MMR coverage is essential CDC MMWR .	Moderate
CDC measles case tracking , 2026	National surveillance	As of 23 April 2026 , CDC reported 1,792 confirmed measles cases in the United States in 2026; full-year 2025 had 2,288 confirmed cases CDC measles cases and outbreaks .	Moderate
WHO Disease Outbreak News: Measles – Region of the Americas , 2025	WHO outbreak update	Recommends high vaccination coverage, rapid response to imported cases, strengthened surveillance, border preparedness, rapid response teams, and vaccine/supply stocks WHO Disease Outbreak News .	Moderate
Reevaluating vitamin A for measles management in high-income settings , 2025	Commentary	Argues for targeted, risk-based vitamin A use in high-income settings because deficiency is rare and excessive intake may cause harm PMC vitamin A commentary .	Low
Navigating Measles Treatment in 2025 , PubMed-indexed review	Review	Reviews ribavirin, MMR PEP, immune globulin, antibiotics, vitamin A, and supplements; emphasizes consultation with healthcare providers and professional organizations PubMed .	Low to Moderate
NACI measles PEP recommendations	National advisory guidance	Susceptible immunocompetent people ≥6 months should receive MMR within 72 hours if no contraindication; immune globulin	Moderate

Study / source	Design	Key findings	GRADE Quality
		should be given only within 6 days; infants <6 months may receive IMIG 0.5 mL/kg up to 15 mL NACI recommendations .	
CDC clinical and infection-control guidance	Public-health clinical guidance	Confirms clinical features, infectious period, IgM/RT-PCR confirmation, isolation for 4 days after rash, airborne precautions, PEP, and vaccine recommendations CDC clinical overview , CDC infection control .	Moderate

2.3 Evidence Gaps and Future Research Needs

Evidence gap	Why it matters	Priority research need	GRADE certainty of current evidence
No approved antiviral therapy for acute measles	Severe measles, immunocompromised infection, and outbreaks require better treatment options.	Randomized or pragmatic trials of antivirals such as ribavirin, interferon- α , or newer agents in outbreak settings.	Low
Limited high-quality PEP evidence in pregnant and immunocompromised populations	These groups have higher complication risk and often cannot receive live vaccine.	Prospective studies evaluating immune globulin dose, timing, effectiveness, and safety.	Low
Vitamin A benefit in high-income settings is uncertain	Universal high-dose use may offer limited benefit where deficiency is rare and may increase toxicity risk.	Risk-stratified trials or comparative effectiveness studies in hospitalized, malnourished, deficient, and otherwise healthy children.	Low to Moderate
Operational best approach to targeted/selective campaigns	Countries need efficient strategies to close immunity gaps with limited resources.	Implementation studies comparing targeted, selective, and nonselective campaign strategies.	Moderate
Surveillance sensitivity and molecular epidemiology gaps	Elimination verification depends on high-quality	Better subnational surveillance-performance metrics, genomic tracking,	Moderate

Evidence gap	Why it matters	Priority research need	GRADE certainty of current evidence
	surveillance and source classification.	and outbreak reconstruction methods.	
Youth engagement in measles guideline development	WHO youth-engagement case study identified is about abortion care, not measles.	If relevant, WHO could develop measles-specific youth/community engagement guidance for vaccine confidence and outbreak communication.	Very Low

2.4 Updated Recommendations Based on New Evidence

Original recommendation	New evidence impact	Updated recommendation
Maintain high measles vaccine coverage.	WHO IA2030 reiterates need for 95% two-dose coverage ; CDC 2025–2026 data show substantial resurgence where immunity gaps persist WHO IA2030 PDF , CDC measles cases and outbreaks .	Strongly maintain or restore ≥95% two-dose MCV/MMR coverage at national and subnational levels.
Use outbreak vaccination and response measures.	WHO 2025 outbreak update stresses rapid response, surveillance, vaccination access for travellers and vulnerable populations, and vaccine/supply stocks WHO Disease Outbreak News .	Activate rapid response teams immediately for suspected outbreaks; conduct contact tracing, vaccination, PEP, and targeted campaigns based on risk assessment.
Confirm measles through surveillance and laboratory testing.	CDC guidance reinforces IgM plus RT-PCR confirmation; WHO lab manual supports diagnosis and molecular characterization CDC infection control , WHO manual page .	Collect serum for measles IgM and respiratory/throat/nasopharyngeal or urine specimens for RT-PCR as early as possible; obtain molecular data during outbreaks to track genotype/source.
Isolate measles cases.	CDC and WHO consistently define infectiousness around rash onset; CDC recommends isolation for 4 days after rash and airborne precautions CDC	Use airborne precautions immediately for suspected measles and isolate through day 4 after rash onset; consider longer precautions in severely immunocompromised patients.

Original recommendation	New evidence impact	Updated recommendation
	clinical overview , WHO Surveillance Standards PDF .	
Use vitamin A in children with measles.	2026 systematic review supports vitamin A as the only widely recommended therapy, while 2025 commentary urges caution against universal unsupervised use in high-income settings PubMed , PMC vitamin A commentary .	Give age-based vitamin A under clinician supervision to children with measles, prioritizing severe disease, hospitalization, malnutrition, or deficiency risk; avoid unsupervised or repeated dosing because of toxicity.
Consider antivirals in severe cases.	Evidence for ribavirin/interferon- α is limited to small trials and case reports; no routine approved antiviral exists PubMed , PMC updated literature review .	Do not use antivirals routinely outside specialist consultation, compassionate-use contexts, or clinical trials.
Provide PEP to susceptible exposed contacts.	2025 systematic review shows potentially high effectiveness but low-quality evidence; CDC/ACOG/NACI provide practical timing PubMed PEP systematic review , CDC vaccine recommendations , ACOG .	Offer MMR within 72 hours to eligible susceptible contacts; offer immune globulin within 6 days to infants, pregnant susceptible people, severely immunocompromised people, or others for whom vaccine is contraindicated; never administer MMR and immune globulin simultaneously.
Verify elimination using interruption of endemic transmission.	WHO regional guidance consistently uses ≥ 36 months of interrupted endemic transmission under high-quality surveillance and defines re-established transmission as ≥ 12 months of uninterrupted transmission after elimination WHO IRIS PDF , WHO EMRO PDF .	Retain the 36-month verification standard and strengthen subnational surveillance, molecular epidemiology, and vaccination-coverage documentation.

Practical Clinical Algorithm

1. **Suspect measles immediately** in any patient with fever and generalized maculopapular rash, especially with cough, coryza, conjunctivitis, Koplik spots, travel, outbreak exposure, or under-immunization.
2. **Mask and isolate immediately**; use airborne precautions in healthcare settings.
3. **Notify public health urgently**; measles is an outbreak-prone disease requiring rapid response.
4. **Test** with measles IgM and RT-PCR; collect specimens early and support genotyping during outbreaks.

5. Treat supportively:

- Hydration, fever management, nutrition, treatment of bacterial complications if present.
- Vitamin A for children, especially severe/hospitalized/malnourished/deficiency-risk cases, using age-based dosing under supervision.

6. Manage contacts:

- Determine presumptive immunity.
- Give **MMR within 72 hours** if eligible.
- Give **immune globulin within 6 days** if high-risk or vaccine contraindicated.
- Do not co-administer MMR and immune globulin.

7. Control outbreak:

- Contact tracing, quarantine/exclusion as directed by public health, targeted vaccination, risk communication, and surveillance enhancement.

8. Document for elimination:

- Classify each case by final classification and source.
- Analyze epidemiology and surveillance performance subnationally.
- Maintain evidence for ≥ 36 months of interrupted endemic transmission.